3. ANALYTICAL GEOMETRY

Exercise 3.1

1. $x^2 - 2x - 6y + 10 = 0$
2. $x^2 + y^2 - 6x + 4y - 3 = 0$
3. $3x^2 + 3y^2 - 4x - 14y + 15 = 0$
4. $\left(\frac{15}{2},\ 0\right)$
5. $2x - 3y + 21 = 0$

Exercise 3.2

1. Angle between the line is 45°
2. 4 units
4. $a = 5$
5. $y = 1500x + 100000$, cost of 95 TV sets ₹ 2,42,500

Exercise 3.3

1. $a = 6,\ c = 6$
2. $2x - y + 2 = 0$ and $6x - 2y + 1 = 0$
3. $2x + 3y - 1 = 0$ and $2x + 3y - 2 = 0$
4. $\theta = \tan^{-1}(7)$

Exercise 3.4

1.(i) $x^2 + y^2 - 6x - 10y + 9 = 0$
(ii) $x^2 + y^2 - 4 = 0$

2.(i) $C(0, 0),\quad r = 4$
(ii) $C(11, 2),\quad r = 10$
(iii) $C\left(\frac{-2}{5}, \frac{4}{5}\right),\ r = 2$
(iv) $C\left(\frac{3}{2}, \frac{3}{2}\right),\ r = \frac{5}{\sqrt{2}}$

3. $x^2 + y^2 + 6x + 4y - 51 = 0$
4. $x^2 + y^2 - 4x - 6y + 11 = 0$
5. $x^2 + y^2 - 5x - 2y + 1 = 0$
6. $x^2 + y^2 - x - y = 0$
7. $x^2 + y^2 - 16x + 4y - 32 = 0$
8. $x^2 + y^2 - 2x - 12y + 27 = 0$
9. $x^2 + y^2 = 9$

Exercise 3.5

1. $x + 2 = 0$
2. R lies inside, P lies outside and Q lies on the circle.
3. $\sqrt{20}$ units
4. $p = \pm 20$.

Exercise 3.6

1. $9x^2 + 16y^2 + 24xy + 34x + 112y + 121 = 0$
2. Length of latus rectum = 1, Focus $F\left(\frac{1}{4},\ 0\right)$

11_11th_BM-STAT_Answer-EM.indd 259 21-04-2020 12:08:50 PM